Xusen GUO, Ph.D. Candidate

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Research Interests:
LLM Agents, Multi-Agent Cooperation, Autonomous Driving, Urban Sensing, and Urban Simulation.

Education

2023.09 – Now	**Ph.D., Intelligent Transportation, HKUST (Guangzhou)** Research Topics: *LLM-based multi-agent, ITS, Urban simulation, Urban sensing.* Supervisor: Prof. Yuxuan LIANG, Prof. Hai YANG
2017.08 – 2020.07	**Master, Computer Science, Sun Yat-Sen University** Thesis: *3D Streaming-based Object Detection and Tracking in Autonomous Driving.* Supervisor: Prof. Kai HUANG
2013.08 – 2017.07	**Bachelor, Material Chemistry, Sun Yat-Sen University** Thesis: *Molecular sieve structure prediction based on computer simulation.*

Research Publications

Journal Articles

[1] **X. Guo**, Q. Zhang, J. Jiang, M. Peng, M. Zhu, and H. F. Yang, "Towards explainable traffic flow prediction with large language models," *Communications in Transportation Research*, vol. 4, p. 100 150, 2024, ISSN: 2772-4247. DOI: https://doi.org/10.1016/j.commtr.2024.100150.

[2] **X. Guo**, X. Yang, M. Peng, H. Lu, M. Zhu, and H. Yang, "Automating traffic model enhancement with ai research agent," *Transportation Research Part C: Emerging Technologies*, vol. 178, p. 105 187, 2025, ISSN: 0968-090X. DOI: https://doi.org/10.1016/j.trc.2025.105187.

[3] M. Peng, **X. Guo**, X. Chen, K. Chen, M. Zhu, et al., "LC-LLM: Explainable lane-change intention and trajectory predictions with large language models," *Communications in Transportation Research*, vol. 5, p. 100 170, 2025, ISSN: 2772-4247. DOI: https://doi.org/10.1016/j.commtr.2025.100170.

[4] M. Peng, K. Chen, **X. Guo**, Q. Zhang, H. Zhong, M. Zhu, et al., "Diffusion models for intelligent transportation systems: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 12, pp. 21 526–21 543, 2025. DOI: 10.1109/TITS.2025.3613178.

[5] Z. Shi, **X. Guo**, H. Lu, M. Peng, et al., "Coordinated pandemic control with large language model agents as policymaking assistants," *arXiv preprint arXiv:2601.09264*, 2026.

[6] M. Peng, Y. Xie, **X. Guo**, R. Yao, H. Yang, and J. Ma, "Ld-scene: LLM-guided diffusion for controllable generation of adversarial safety-critical driving scenarios," *Transportation Research Part C: Emerging Technologies*, 2026.

[7] J. Hu, **X. Guo**, J. Chen, X. Liu, J. Qu, et al., "Zeolite LTA structure generation by coordination sequence and vertex symbol," *Microporous and Mesoporous Materials*, vol. 298, p. 110 050, 2020, ISSN: 1387-1811. DOI: https://doi.org/10.1016/j.micromeso.2020.110050.

[8] L. Cheng, N. Liu, **X. Guo**, Y. Shen, Z. Meng, K. Huang, et al., "A novel rain removal approach for outdoor dynamic vision sensor event videos," *Frontiers in Neurorobotics*, vol. 16, p. 928 707, 2022.

[9] R. Wang, D. Wang, D. Kang, **X. Guo**, C. Guo, et al., "An artificial intelligent platform for live cell identification and the detection of cross-contamination," *Annals of Translational Medicine*, vol. 8, no. 11, p. 697, 2020.

Conference Proceedings

[1] **X. Guo**, M. Peng, H. Lu, H. Yang, J. Ma, and Y. Liang, “Language-grounded multi-agent planning for personalized and fair participatory urban sensing,” in *arXiv preprint arXiv:2603.24014. ACL 2026 (under review).*

[2] **X. Guo**, M. Peng, X. Hao, X. Zou, Y. Liang, et al., “Agentsense: LLMs empower generalizable and explainable web-based participatory urban sensing,” in *The ACM Web Conference (WWW), Oral*, 2026.

[3] X. Hao, G. Li, D. Wu, **X. Guo**, Y. Liang, et al., “Enhancing ride-hailing forecasting at didi with multi-view geospatial representation learning from the web,” in *The ACM Web Conference (WWW)*, 2026.

[4] **X. Guo**, M. Peng, X. Hao, X. Zou, Y. Liang, et al., “Towards responsible and explainable traffic flow prediction with large language models,” in *Transportation Research Board, (Oral)*, 2025.

[5] M. Peng, R. Yao, **X. Guo**, Y. Xie, X. Chen, and J. Ma, “Safety-critical traffic simulation with guided latent diffusion model,” in *2025 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC). IEEE*, 2025.

[6] M. Peng, R. Yao, **X. Guo**, and J. Ma, “Nuplan-r: A closed-loop planning benchmark for autonomous driving via reactive multi-agent simulation,” 2026.

[7] **X. Guo**, J. Gu, S. Guo, Z. Xu, H. Kai, et al., “3d object detection and tracking based on streaming data,” in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 8376–8382. DOI: 10.1109/ICRA40945.2020.9197183.

Industrial Experience

2023.05 – 2023.08 — **Banma.ai.** Senior Perception Algorithm Engineer, responsible for developing and deploying obstacle perception algorithms for *Autonomous Parking Assistance systems*, including leading the development of fisheye-based obstacle detection models with BEV projection and optimizing them for real-time deployment on onboard chips with ultra-low resource consumption (10 FLOPs).

2020.09 – 2023.05 — **Waytous Technology.** Perception Algorithm Engineer, responsible for developing vehicle-road cooperative perception algorithms for mining scenarios, including designing a unified *vehicle-road-cloud post-fusion framework* with hot-swapping capability, developing real-time *roadside perception algorithms* (object detection, tracking, and trajectory prediction), building an integrated training and deployment platform that improved 3D detection accuracy from 84.6% to 96.2% at 50Hz, and implementing a LIDAR-based truck *load measurement system* with 98.6% accuracy.

Awards and Achievements

- 2026 — *ACM SpatialDI 2026 Outstanding Academic Poster Award*.
- 2017-2019 — *Second-level Excellent Graduate Student Scholarship*, Sun Yat-Sen University.
- 2018 — *AIIA Cup: Medical Artificial Intelligence Challenge First Prize*.
- 2017 — *Graduate Student Recommended Exemption*, Sun Yat-Sen University.
- 2016 — *First Prize of Undergraduate Research Competition*, Sun Yat-Sen University.
- 2014-2016 — *Second-level Excellent Student Scholarship*, Sun Yat-Sen University.